

7. Project Documentation

OptiCrop – Smart Agricultural Production Optimization Engine

Software Used

- Python
- Flask
- Werkzeug
- Pandas
- NumPy
- SciPy
- Matplotlib
- Seaborn
- Scikit-learn
- Joblib
- ReportLab

Installation Steps

1. Clone the repository.
2. Open the project in Visual Studio Code.
3. Install dependencies.

`pip install -r requirements.txt`

4. Train the model (once).

`python model_training.py`

5. Run the application.

`python app.py`

6. Open the browser.

`http://127.0.0.1:5000/`

User Guide

Step 1 — Create an account

Go to /register, enter your name, email, password, confirm password, and select a role (Farmer, Researcher, Agribusiness, or Government).

Step 2 — Log in

Go to /login and sign in.

Step 3

On the Overview page, use the sidebar to reach Smart Recommendation.

Step 4

Enter Nitrogen (N), Phosphorous (P), Potassium (K), Temperature (°C), Humidity (%), Soil pH and Rainfall (mm).

Step 5

Click “Recommend crop” (or “Fill sample values” to auto-fill a test example). View the recommended crop, its confidence percentage, and the auto-generated report/recommendations. Click “Download PDF report” to save a styled PDF of the result.

Step 6

Optionally, go to Suitability Assessment to check a specific crop against your field conditions, or Crop Library to browse crop details.

Step 7

Go to Prediction History to view all past recommendations. Type a crop name to search; click “Export CSV” to download exactly the rows currently shown (leave the search empty to export everything).

Project Structure

- database/
- dataset/
- model/
- notebooks/
- src/
- static/
- templates/
- app.py
- auth.py
- database.py
- model_training.py
- README.md
- requirements.txt
- utils.py

Conclusion

This documentation successfully demonstrates the application of Machine Learning in agriculture by providing intelligent crop recommendations based on soil and environmental conditions. The project promotes data-driven decision-making, sustainable farming practices, and efficient resource utilization through a user-friendly web application.